Kruskal’s Minimum Spanning Tree

Topic: Greedy File: g4.pdf

PROBLEM

For a connected, weighted, undirected graph, find a spanning tree with the minimum total edge weight.

GREEDY CHOICE

Sort edges by increasing weight and add an edge only when it connects two different components.

ALGORITHM

Sort edges. Initialize a disjoint-set structure. For each edge (u,v), if find(u) != find(v), add it and union the components. Stop after V - 1 edges.

EXAMPLE

For edges AB:1, BC:2, AC:3, choose AB and BC. AC is skipped because it would make a cycle.

COMPLEXITY

Sorting costs O(E log E). Union-Find operations are almost constant amortized time, so sorting dominates.

REQUIREMENT

For a disconnected graph, the same procedure produces a minimum spanning forest.